

provedeme i součin na pravé straně a dostáváme:

$$\begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}, \text{ nyní vyřešíme SLR}$$

$$\text{I. } 4a - b + 2c = 2$$

$$\text{II. } -a + 4b + 2c = 4 \Rightarrow a = 4b + 2c - 4$$

$$\text{III. } 2a + 2b + 6c = 6$$

$$\text{I. } 16b + 8c - 16 - b + 2c = 2$$

$$\text{III. } 8b + 4c - 8 + 2b + 6c = 6$$

$$\text{I. } 15b + 10c = 18$$

$$\text{III. } 10b + 10c = 14$$

) I. - III.

$$\left. \begin{array}{l} \text{I. } 15b + 10c = 18 \\ \text{III. } 10b + 10c = 14 \end{array} \right\} \Rightarrow \left. \begin{array}{l} \text{I. } 10c = 6 \Rightarrow c = \frac{6}{10} = \frac{3}{5} \\ \text{II. } 10c = 6 \Rightarrow c = \frac{6}{10} = \frac{3}{5} \end{array} \right\} =$$

$$\Rightarrow a = \frac{16}{5} + \frac{6}{5} - \frac{20}{5} = \frac{22-20}{5} = \frac{2}{5}$$

$$a = \frac{2}{5}$$

$$b = \frac{4}{5}$$

$$c = \frac{3}{5}$$

$$\boxed{\text{ŘEŠENÍ: } M = \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5}}$$